

Reimagining feedback through generative AI in engineering education

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ABSTRACT

The rapid integration of Generative AI (GenAI) into higher education presents both opportunities and systemic challenges, particularly in domains where feedback is central to learning. This study investigates the capacity of a large language model to generate formative feedback for student-created UML diagrams in a university software engineering course. Across two cohorts ($N = 262$), AI-generated, teacher-generated, and no-feedback conditions were compared, analyzing student perceptions, learning outcomes, and grading reliability. Results show that while students rated GenAI feedback as beneficial and often comparable to human comments, teacher feedback remained more effective in supporting performance gains, especially in complex modeling tasks. Linguistic analysis further revealed that GenAI feedback was more repetitive and less pedagogically rich than human feedback. Beyond these course-level findings, the study highlights broader implications for higher education systems. GenAI feedback represents not just a pedagogical tool but a cognitive partner that can reshape assessment models, curriculum design, and faculty roles. Its scalable nature offers potential to democratize access to high-quality formative feedback, while also raising equity, accountability, and policy challenges at institutional and sectoral levels. By situating empirical results into this broader frame, the study argues that GenAI is catalyzing a paradigm shift toward new systems of learning, where feedback becomes systemic, scalable, and embedded in the very structure of higher education.

1. Introduction

The introduction of ChatGPT has opened possibilities to integrate accessible Generative Artificial Intelligence (GenAI) and Large Language Models (LLMs) into education. These models have proven their capabilities across tasks such as code generation (Liu et al., 2023), test automation (Zimmermann & Koziol, 2023), and assessment (Gao et al., 2024; Henkel et al., 2024). When prompted accordingly, GenAI can perform a wide range of functions, presenting opportunities to support educators in time-intensive tasks and enhance student learning experience. One domain where this potential is especially relevant is diagrammatic modeling, particularly the design and evaluation of Unified Modeling Language (UML) diagrams. UML diagrams are considered a fundamental part of software engineering, with their ability to visually describe the structural aspects of software systems (Ozkaya & Erata, 2020). Designing a UML diagram requires the students to translate abstract system requirements into structured visual representations, which makes these tasks high in cognitive load and involves trade-offs in design, structure, and semantics. Diagrammatic modeling tasks, such as

UML design, database schemas, and object-oriented structures, demand high levels of abstraction and logical precision, making them pedagogically valuable due to their semantic complexity yet assessment-intensive nature. Unlike programming or writing tasks, UML diagrams are open-ended, with no single correct solution, and typically demand expert evaluation to assess correctness, structural consistency, and design logic. This makes timely and high-quality feedback difficult to scale, particularly in high-enrollment or online courses where individualized instructor feedback is impossible.

From a pedagogical perspective, feedback on UML and other diagrammatic modeling tasks represents a long-standing challenge in engineering education. Such tasks are inherently open-ended, allow for multiple valid solutions, and require evaluative judgment that goes beyond surface-level correctness. Prior research has shown that while formative feedback is critical for supporting conceptual understanding and design reasoning, providing timely, individualized, and pedagogically rich feedback on complex visual models is difficult to scale, particularly in large or online courses (Boud & Dawson, 2023; Winstone & Carless, 2019).

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Existing approaches to automated UML assessment have primarily focused on rule-based systems or machine learning techniques aimed at grading accuracy rather than learning support (Beck et al., 2015; Bergström et al., 2022; Correia et al., 2017). Although these systems can detect syntactic or structural errors, they often struggle to deliver explanatory, design-oriented, or reflective feedback that supports higher-order learning processes. As a result, a persistent gap remains between scalable automated assessment and pedagogically meaningful formative feedback in diagrammatic modeling contexts.

GenAI tools are increasingly integrated into educational practices, like designing inquiry-based learning lesson plans (Moundridou et al., 2024), generating codes (Liu et al., 2023), or automating feedback across various domains (Gao et al., 2024; Henkel et al., 2024; Pecuchova & Drlik, 2025). GenAI can improve student motivation and the learning process by offering immediate feedback (Kasneji et al., 2023). Against this backdrop, GenAI has emerged as a potential response to the scalability challenges of formative feedback, rather than merely as a tool for automation or efficiency. Unlike earlier rule-based or task-specific systems, LLMs offer the ability to generate context-sensitive, natural-language feedback that may address not only correctness but also reasoning, design choices, and reflection. In programming education, GenAI automated assessment helps move beyond traditional systems that provided only limited feedback or relied on predefined templates (Zhang et al., 2024). They have been used to offer detailed explanations for student mistakes or provide single-sentence hints in a conversational manner (Heickal & Lan, 2024). These models have led to the development of various programming feedback tools that are being deployed in classrooms and offer quality feedback through iterative improvement (Prather et al., 2024). Automatization of feedback can be beneficial for high-enrollment courses, where it can adjust the learning experience to each student's progress and learning style (Y. Dai et al., 2023; Zhang et al., 2024). These benefits extend beyond programming, where, for example, in writing instruction, GenAI has produced feedback rated as more readable and consistent than human-generated feedback, though often less precise (Biber et al., 2011; Fleckenstein et al., 2023). Dai et al. (2024) showed that GenAI-generated feedback outperforms human instructors because they often struggle to deliver quality feedback, mainly in large classes consistently. However, studies highlight GenAI's limitations in handling domain-specific language, producing hallucinated content, and failing to capture conceptual misunderstandings without careful prompt design. These shortcomings underscore the need for hybrid systems (Azaria et al., 2024; Phung et al., 2024), where human supervision supports GenAI feedback generation (Tuomisto & Harjuma, 2025).

More recently, GenAI has begun to move from standalone applications toward infrastructure integration in education. This implies a more integrated and scalable role within existing systems, moving beyond human interaction (Kinder et al., 2025). Feedback generation is being embedded into educational platforms and learning management systems (LMS) through APIs, transforming GenAI into a background service capable of producing on-demand, context-aware responses to requests (Pereira Cipriano et al., 2024). This shift supports modular system design, allowing for scalable feedback that is maintainable, configurable, and extensible across courses and existing educational platforms, and reusability (Amiri & Islam, 2025). One of the most significant ways GenAI becomes an infrastructural element is by overcoming the traditional challenges of providing personalized and adaptive feedback at scale (Kinder et al., 2025). When combined with structured input data, such as student-submitted UML diagrams, GenAI models may function not only as evaluators but as feedback systems capable of supporting large cohorts with minimal teacher involvement. This changes the capacity of feedback delivery, making it a systemic, pervasive capability rather than a bottleneck dependent on human availability (Kinder et al., 2025). This transformation reflects a broader paradigm shift in education systems, where GenAI is positioned not as a supplementary tool but as an instructional component capable of reshaping assessment,

feedback, and instructional design (Sandu et al., 2024; Wang et al., 2024). Understanding the suggestions of this shift requires empirical studies that evaluate GenAI's educational value in complex, structured tasks, such as UML modeling.

Despite growing interest in the use of GenAI for automated assessment and grading in software engineering education, empirical evidence on its capacity to provide pedagogically meaningful formative feedback for complex, diagrammatic modeling tasks remains limited. In particular, little is known about how GenAI-generated feedback compares to human instructor feedback in terms of feedback types, instructional richness, and its impact on student learning outcomes beyond surface-level performance. Addressing this gap is essential for understanding whether GenAI can function as a learning-oriented feedback mechanism rather than merely an evaluative or grading tool. The results achieved in evaluation of GenAI feedback to programming tasks (Koutchme et al., 2025; Lai & Lin, 2025; Sandu et al., 2024) show a promising direction and offer potential to address in this paper.

This study investigates the quality and usefulness of GenAI-generated feedback for UML design assignments in a university software engineering course. Specifically, the aim is to evaluate how well GenAI can identify structural or semantic issues in student-submitted UML diagrams, and how students interpret and value this feedback. Unlike prior work focused primarily on grading, the goal of this study is to assess GenAI's capacity to support learning through formative feedback in structurally complex, diagrammatic modeling tasks.

This study contributes a perspective by demonstrating how GenAI, when deployed as an embedded feedback infrastructure, can extend beyond isolated tutoring scenarios and function as a scalable cognitive partner in educational ecosystems. What makes this approach distinctive is the integration of GenAI into diagrammatic modeling, a domain traditionally reliant on expert interpretation, and also the systemic configuration of feedback as a service-level educational component. The study explores how prompt design, input structure, and task complexity interact with GenAI's behavior, offering insights into the stability and robustness of AI-generated feedback across varied instructional contexts. This has direct implications for building hybrid teacher-AI assessment models, where educators focus on mentoring, nuance, and higher-order guidance, while GenAI supports high-frequency formative feedback loops.

By embedding GenAI into the feedback architecture of complex, open-ended tasks, the research moves toward a model of feedback-abundant learning that may help overcome the long-standing trade-off between cognitive load and instructional scalability. The findings support a shift in educational systems from viewing AI as an auxiliary tool toward suggesting how feedback may increasingly be conceptualized as an infrastructural capability within hybrid human-AI educational systems, a perspective aligned with broader calls to rethink curriculum and pedagogy in the age of cognitive technologies. In doing so, this work offers both empirical grounding and forward-looking extrapolation into how GenAI can support scalable, personalized, and pedagogically meaningful assessment systems in higher education.

The paper is organized as follows. The related work on integrating GenAI into education is summarized in Section 2. Section 3 details the dataset used and the methods to achieve the aim of the experiment. Section 4 presents the results of producing feedback using GenAI and evaluating the quality of the generated feedback. Section 5 provides a discussion of the results and implications for education. Subsequently, the conclusions and ideas for future research are presented.

2. Related work

GenAI offers new opportunities to enhance feedback delivery in computing education. Across various tasks, its ability to generate timely, context-sensitive feedback has been examined to help students identify errors, understand concepts, and develop self-directed learning strategies. Frankford et al. (2024) examined the GPT-3.5-powered AI-Tutor

designed to give feedback to students' coding assignments. The model was set to guide students to solve the coding issues without providing direct solutions. They identified two usage patterns by students that can help in designing AI-powered educational systems. The first pattern involved students using continuous prompting from the AI-Tutor. While this was beneficial for students seeking guidance, there was a rising concern about over-reliance on the AI-Tutor. The second pattern was oriented towards an iterative learning approach, where the students relied heavily on the test feedback to guide their solution. While students appreciated the timely and scalable feedback, the system also produced occasional generic responses and required careful instructional framing. Messer et al. (2024) noted in a systematic review of automatic feedback systems that most studies lack human evaluation of AI-generated feedback, limiting the understanding of its instructional quality.

Other studies focused on more specific tasks. Ahmed et al. (2019) used targeted examples for compilation errors and compared the time taken to repair these errors with and without feedback. On average, the feedback improved the time necessary to repair the bugs. Pădurean et al. (2025) argue that students mainly use the trial-and-error methods to resolve coding issues, while the ability to identify and hypothesize the cause of errors is crucial. They used GenAI to generate buggy assignments from a problem description and verify them using a test suite. The results showed that GenAI generates comparable assignments to teachers, and both GPT-4o and GPT-3.5 were able to generate diverse sets of exercises with varying difficulty. This indicates that for the GenAI models to produce quality feedback, they have to be able to create faulty assignments that teach students to identify issues and improve critical thinking. Koutcheme et al. (2025) compared open-source and proprietary LLMs, evaluating their ability to generate and assess feedback. Proprietary models showed the highest performance across all the criteria examined. The authors identified quality gaps by identifying irrelevant issues. However, most of the time, they produced understandable feedback. The results showed potential in the integration of open-source LLMs into feedback tools. However, some models require a lot of computational power, which can make integration difficult. The results also showed that smaller models could, with careful fine-tuning, be useable for feedback generation. Sandu et al. (2024) and Lai and Lin (2025) extended these findings to curriculum level developments, showing how GenAI could support high-enrollment courses and offer dynamic, personalized feedback. However, concerns remain about over-reliance, hallucinations, and insufficient conceptual depth in GenAI-generated responses.

Providing personalized feedback is a challenging task for teachers. Despite the advancements, studies show that students often find AI-generated feedback less helpful than instructor feedback. Er et al. (2025) found that GenAI models lack contextual sensitivity and need context-specific training to produce helpful feedback. The students perceived the instructors' feedback as more useful, and students receiving AI feedback showed lower improvement in learning outcomes. Similarly, Li et al. (2025) observed that while GenAI can enhance learning outcomes in the short term, it may negatively affect long-term knowledge transfer. This creates the risk of over-reliance on GenAI, as the students trust the feedback without critical thinking. The authors recommended using GenAI to support student engagement, while encouraging other problem-solving assignments without external resources. This way, students can improve their academic performance while maximizing learning benefits. Similar concerns are raised by Sheese et al. (2024), who noted that students often submit minimal prompts, missing deeper conceptual understanding. This raises the requirement of human supervision to maintain feedback quality.

The UML diagrams are used in the analysis and construction of software systems. The quality of the layout is essential to share knowledge about the system between developers. Automated assessment of the designing tasks has been a research goal, particularly for use in online learning settings. Bergström et al. (2022) used machine learning

techniques to build regression models based on features extracted using image processing. The models were trained on human-labeled diagrams using a 5-point Likert scale. Despite limitations of a small dataset, their approach achieved strong alignment with human evaluators, showing the potential for scalable grading. Beck et al. (2015) integrated rule-based feedback for UML diagrams through LMS Moodle. However, their approach was limited by the specification of the hand-coded rules, which struggled with anticipating diverse student errors. Correia et al. (2017) addressed this by proposing an automated diagram assessment tool with enhanced feedback. The proposed tool can distinguish between syntactic and semantic errors and provide specialized feedback. The authors argued the need for conceptual, formative feedback, not just error marking, which would help students understand the logic behind designing UML diagrams.

More recently, GenAI has been explored as an evaluator in visual design contexts. Wang et al. (2024) evaluated the quality of UML use case, class, and sequence diagrams. They created a set of evaluation criteria for GenAI to assess the quality of examined UML diagrams quantitatively. ChatGPT showed the capability to grade UML diagrams with similar performance to teachers. Wang et al. (2024) suggested new directions in integrating GenAI to software engineering education, where GenAI can offer objectivity of the grading process and fairness of scores to some degree. Another point provided was that GenAI can offer valuable insights into students' learning situations and help identify potential weaknesses. Similar results were obtained by Bouali et al. (2025) who compared three GenAI models' capabilities to evaluate UML class diagrams. The authors highlighted that the assessment of UML diagrams is a subjective challenge due to the absence of unique, specific answers. However, a well-designed and configured prompt can help the GenAI model to effectively grade these subjective tasks with similar performance to human evaluators. These studies confirmed the assumption about the capability of GenAI in structured visual tasks. However, it was focused primarily on grading rather than formative, student-facing feedback.

The previous research (Pecuchova & Driik, 2025) evaluated GenAI models for assessing responses to open-ended questions. This research provided one point of view on the possibility of using GenAI models to work with textual data that lacks objective reference answers. UML design tasks share this challenge of subjectivity and structural uncertainty, where multiple correct solutions exist. While existing studies tested GenAI's ability to generate or grade UML diagrams, this study focuses on the quality and perceived usefulness of GenAI-generated feedback for student-created UML diagrams. By comparing GenAI feedback with teacher-produced feedback, randomly combined and evaluated by students, the aim is to introduce a novel perspective on how GenAI can be used not just to automate assessment grading, but also to support learning through formative feedback in visual modeling tasks.

3. Materials & methods

Complex and semantically rich schemas, maps, and diagrams are an essential and irreplaceable part of learning in higher education settings. Especially in engineering, business, or computer science study programs, complex visual modeling tasks such as UML and BPMN diagrams, mind maps, database schemas, object-oriented or architectural design patterns are integral components of learning, which can prove the student's correct understanding of the topic. These tasks engage students in abstraction, system thinking, and semantic reasoning. However, they are simultaneously time-consuming to assess and provide feedback on. Especially in high-enrollment courses, the formative evaluation of such diagrammatic assignments often exceeds the available instructional capacity. As such, missing or superficial feedback frequently causes a system-level issue, reducing the opportunity for iterative improvement and self-regulated learning.

This study builds on the assumption that GenAI can serve as an

infrastructural solution to the challenge of scalable formative feedback. Moving beyond its traditional role in natural language or code generation, GenAI was considered as an autonomous feedback agent embedded into a visual modeling workflow. Student-designed UML diagrams, created using Enterprise Architect, were exported in XML format and submitted to a large language model (GPTo1). The model returned natural-language feedback in Slovak. Unlike template-based systems (Beck et al., 2015; Correia et al., 2017), the feedback was generated in a free-text form, with no constraints regarding tone, style, or pedagogical intention. As a result, these settings allow post-hoc analysis of the model's communicative behavior.

This study employed a mixed-methods, quasi-experimental research design conducted across two academic years in a bachelor-level Software Engineering course. A total of 262 students were divided into three groups: one receiving feedback generated by a generative AI model, one receiving teacher-provided feedback, and a control group receiving no feedback. The design combined quantitative analyses (non-parametric statistical tests, regression models, inter-grader reliability metrics, and linguistic/sentiment analyses) with qualitative categorization of feedback into diagnostic, design-oriented, motivational, and metacognitive types. This multi-cohort, multi-modal approach allowed us to evaluate both the pedagogical quality and systemic implications of AI-generated feedback, capturing not only its impact on student learning outcomes but also its communicative characteristics and alignment with human-provided feedback. The research design aimed to address multiple layers of pedagogical evaluation, from affective tone to technical alignment. It was analyzed how GenAI feedback is perceived by students, what types of comments it tends to generate, and whether it can match the evaluative reliability of human experts. Additionally, the research investigated the downstream effects of receiving GenAI feedback on student learning outcomes, comparing them to those who received instructor feedback or no feedback at all. To systematically explore these aspects, the following research questions were formulated:

RQ1: How can the student-perceived helpfulness and pedagogical value of GenAI-generated feedback be compared to the instructor's feedback across UML modeling tasks?

RQ2: What types of feedback does GenAI produce spontaneously without explicit prompt conditioning, and how does this distribution compare to that of teacher-generated feedback?

RQ3: Does receiving GenAI feedback influence students' modelling skills and improve their performance on subsequent modeling tasks and final exams, relative to students receiving teacher or no feedback?

RQ4: Can GenAI accurately grade visual UML diagrams based on XML representations with agreement levels comparable to expert human graders, and how do different prompt strategies influence this grading performance?

The following steps were applied to answer the research questions:

1. Data Acquisition
 - a. Gathering data from student assignments focused on creating UML diagrams.
 - b. Creating feedback (GenAI and teacher) to students and obtaining their evaluation of the feedback using a five-point Likert scale.
2. Assignment Grading
 - a. Assessment of assignments by two independent graders.
 - b. Evaluation of feedback into four categories (diagnostic, design, metacognitive, motivational).
3. Data preparation
 - a. Creating a data matrix containing all grades and assignments per student.
4. Data analysis and Evaluation
 - a. Feedback analysis using non-parametric tests and linguistic metrics.

- b. Post-hoc sentiment analysis of teacher and GenAI feedback.
- c. GenAI assessment of UML diagrams evaluation.

5. Results Interpretation

- a. Formulation of conclusions, benefits, and limitations of GenAI integration.

The forthcoming subsections describe the dataset structure, GenAI feedback generation pipeline, evaluation procedures, and analysis methods used to address the research questions in detail.

3.1. Dataset

The study was conducted over two consecutive academic years (2023/2024 and 2024/2025) at Constantine the Philosopher University in Nitra, as part of the bachelor's course *Introduction to Software Engineering*. The identical instructional content, assignments, and assessment protocols were used in both years, allowing data to be combined for analysis. A total of 262 students participated in the experiment across both years. Over a 13-week semester, students completed four primary assignments, each targeting a distinct phase of software system design: (1) requirements specification, (2) use case diagram design, (3) class diagram development, and (4) construction of the activity diagram. Each assignment increased in complexity and abstraction. All submissions were exported from Enterprise Architect in XMI/XML format. This format allows for preserving semantics and relationships between modelled elements of the diagram. It ensured consistent, structured, and machine-readable input suitable for both human and AI evaluation. The submission and grading process was managed via the university's LMS (Moodle). Students were divided into three experimental groups to evaluate the effectiveness of different feedback strategies:

- **72 students** received feedback generated by GenAI (GPTo1),
- **39 students** received feedback from a human instructor,
- **151 students** did not receive any feedback (control group).

In addition to the four modeling tasks, students completed six auto-graded theoretical quizzes and a final exam consisting of three UML diagramming tasks (use case, class, and activity). Two human evaluators independently graded these final diagrams and were also assessed by the GenAI model using multiple prompt configurations. All AI-generated feedback was returned in Slovak. This structure allowed for a multi-dimensional analysis of both performance and perception.

3.2. Methods

The key objective of the study was to explore whether LLMs can generate formative feedback on complex visual modeling tasks, specifically UML diagrams, in a manner that is pedagogically relevant and scalable. This study aimed to evaluate the capabilities of generative AI to produce formative feedback and rubric-aligned evaluation for UML modeling tasks in a higher education context. For this reason, a multi-stage methodological pipeline was designed, which consisted of four key components: (1) feedback generation via API-based prompting, (2) scoring and grading of final exam submissions, (3) feedback annotation for research purposes, and (4) data processing and analysis.

3.2.1. Feedback generation process

To simulate scalable formative assessment for diagrammatic tasks, this study integrated GPTo1 into the feedback pipeline via API calls. Throughout the semester, students submitted four assignments via the university's LMS Moodle. Each assignment targeted a distinct phase of software system design: (1) requirements specification, (2) use case diagram design, (3) class diagram development, and (4) construction of the activity diagram. The assignments were designed as open-ended problem scenarios rather than pre-structured templates. Each assignment included a system-level textual description (e.g., of a car rental

system) with specific domain requirements. These open-ended formulations capture the essence of engineering education, where effective problem-solving combines technical skills with the interpretation of unclear or incomplete requirements. In this context, students were ex-

pected to identify appropriate abstractions and represent them using UML diagrams.

grade. The code snippet below illustrates the logic used for dynamic prompt generation.

Prompt 1 – Prompt for API-based Feedback Generation.

```
for index, row in df.iterrows():
    prompt_payload = {
        "instruction": (
            "You are an academic evaluator for a university course in Software Engineering."
            "The student has submitted a UML model in XML format, created as a response to an open-ended system design assignment."
            "Your task is to assess the technical and conceptual quality of the submission."
            "Provide constructive feedback in Slovak that helps the student understand what was done well and what could be improved."
            "You may also include suggestions or guiding questions that encourage the student to reflect on their design rationale and consider possible refinements."
        ),
        "input": {
            "assignment_type": row["assignment_type"],
            "assignment_description": row["assignment_description"],
            "submission_xml": row["submission_xml"]
        }
    }
```

pected to identify appropriate abstractions and represent them using UML diagrams.

For GenAI feedback generation, these original descriptions of assignments were included in the design of the AI prompt. The aim was not to test the model's ability to follow rigid scripts, but to evaluate its performance under conditions resembling authentic classroom instruction. The model had to infer expectations from the assignment narrative and align them with the structure of the submitted UML diagrams, assessing both technical completeness and design coherence.

Students submitted their work as .eapx files, proprietary project files created in Enterprise Architect. These files contained the full project structure, including diagram layout and modeling elements. To enable input into the language model, a manual conversion step was required. Each .eapx file was opened in Enterprise Architect and exported to an XML/JSON format, which preserved the semantic structure of the diagrams in a machine-readable representation. Visual representations (e.g., images or PDFs) in early assignments were also considered. However, image-based inputs were excluded from the AI pipeline due to their inconsistent layout, poor formatting, and the lack of semantic structure. As a result, the XML-based representation was selected as the primary input for generative processing, allowing the semantic content of diagrams (actors, classes, relationships, etc.) to be processed in a format readable by the GPTo1 model. Each XML file was paired with its corresponding assignment.

Although the prompt did not explicitly instruct the model to follow a particular feedback style, it was carefully constructed to emphasize *understanding*, *design reasoning*, and *semantic accuracy*, thus implicitly encouraging the model to move beyond binary correctness and address conceptual development. In this way, the feedback generation process was aligned with contemporary views on GenAI as a cognitive partner in education, which supports not only performance correction but also deeper learning through iterative self-evaluation and model-based scaffolding (Huang et al., 2022). The feedback generation was handled by a Python script that iterated through a structured dataset, where each row represented one student submission. The dataset contained the following columns *student_id*, *assignment_type*, *assignment_description*, and *submission_xml*. For every record, a JSON-formatted prompt was dynamically constructed and submitted to the GPTo1 model via an API. The model's task was to evaluate the student submission based on the instructional context provided and return feedback in natural language (Slovak) along with a rubric-aligned score and a mapped academic

The prompt was intentionally open-ended to simulate a realistic instructional setting and allow the model to demonstrate emergent feedback behaviors. To preserve the authenticity of the AI-generated output, no post-editing, filtering, or stylistic correction was applied. In line with the study's cognitive design goals, the feedback was not limited to correctness. Ideally, it would also prompt students to reflect on their design decisions, question structural assumptions, and revise their models—thus fostering self-regulated learning (SRL). While the model was not explicitly instructed to produce metacognitive or motivational feedback, the unstructured format enabled its occasional emergence, which was subsequently annotated in post-hoc analysis.

3.2.2. Feedback annotation

To better understand the communicative role of feedback in the instructional process, a post-hoc qualitative annotation process was conducted for each feedback, whether authored by a human instructor or generated by GenAI. This process aimed to categorize each feedback comment into functional pedagogical types, allowing us to evaluate not only the correctness of feedback but also its intended instructional role in the learning process. The categorization scheme was inspired by recent pedagogical literature emphasizing the formative and dialogic role of feedback in higher education (Boud & Dawson, 2023; N. E. Winstone & Boud, 2022). In addition to its corrective functions, feedback is increasingly understood as a means of promoting reflection, engagement, and design awareness, particularly in complex tasks such as visual modeling (N. Winstone & Carless, 2019). Drawing from established pedagogical frameworks, and in alignment with the cognitive goals of the study, a four-category scheme, adapted to the context of diagram modeling tasks, was adopted:

- (1) Diagnostic — Feedback that identifies errors, incomplete elements, or violations of formal diagram syntax or semantics. This includes both surface-level and conceptual mistakes, and it is foundational for student awareness of misunderstandings.
- (2) Design-Oriented — Feedback that comments on modeling strategy, structure, or design decisions without labeling them as errors. For example, advice on class cohesion, diagram readability, or abstraction. This feedback supports deeper engagement with modeling principles.

- (3) Motivational — Feedback that encourages students through praise, acknowledgment of effort, or expressions of confidence in their abilities. While not evaluative in content, such feedback supports affective aspects of learning.
- (4) Metacognitive — Feedback that prompts students to reflect on their own thinking or modeling decisions, such as “Why did you choose this structure?” or “Consider whether this relationship type matches the intended behavior.”

Each feedback item was assigned to a single category based on its dominant communicative intent. Annotation was performed manually by one evaluator familiar with the course content and the educational context. While multi-label annotation could have offered additional nuance, a single-label scheme was selected to maintain consistency and analytical clarity in downstream comparisons. This categorization enabled a structured comparison of how feedback types varied across assignments and feedback sources. Of particular interest was the degree to which GenAI-generated comments exhibited higher-order pedagogical characteristics, such as metacognitive prompting or design-oriented guidance, even though the model was not explicitly instructed to adopt these styles.

3.2.3. Grading of final exam submissions

The final exam was designed to assess students' modeling proficiency in a summative context, following the sequence and complexity progression used throughout the course. Each student was required to submit three UML diagrams - use case, class, and activity diagram. To establish a benchmark, each student submission was independently assessed by two independent graders. Unweighted Cohen's Kappa was used to evaluate the alignment between two teachers across all three tasks. The classification of agreement in the case of various metrics in this study was as follows:

- poor (<0.00),
- fair (0.00 – 0.20),
- moderate (0.41 – 0.60),
- substantial (0.61 – 0.80),

perfect agreement that would be appropriate, but the unweighted Cohen's Kappa does not consider the degree of difference between grades. UML diagrams assessment is an abstract task that is also hard to evaluate objectively, even with predefined subtasks. Because of this reason, quadratic weighted Cohen's Kappa (QWK) was used to compare the grader's agreement across the tasks. Almost perfect agreement was identified for all tasks (QWK use case = 0.91; QWK class = 0.92; QWK activity = 0.95). Based on this finding, it was possible to compare the output of the GenAI model assessment with two human graders and obtain relevant results.

In parallel, a targeted evaluation of GenAI's grading capabilities was conducted by designing and deploying three distinct prompts for each of the three UML tasks:

- Prompt 1 tested how the model performs with minimal structured input and indirect evaluation criteria.
- Prompt 2 leveraged structured inputs for more rigorous rubric-based evaluation, aligning with the course's formal expectations.
- Prompt 3 emulated a more granular and realistic grading protocol, suitable for institutional standardization and fairness considerations.

These prompts differed in structure, input modality, and the degree of rubric specificity. The goal was to explore how prompt design influences the accuracy and behavior of the language model when applied to abstract diagrammatic content. This experimental design was also motivated by the absence of a fine-tuned or supervised model. Given that the system operated in a zero-shot configuration, the aim was to explore whether changes in input formulation would impact grading consistency and quality. Each model evaluation was triggered through an API call. A Python script iterated through each student submission, and depending on the prompt condition, either an image (Prompt 1) or XML representation (Prompt 2 and 3) of the diagram was used as input to the model. The prompts were submitted via API to the GPTo1 model and can be summarized as follows.

Prompt 1 - Tests how the model performs with minimal structured input and indirect evaluation criteria.

```
ucd_prompt1 = {
    "model": "gpt-ol",
    "input": {
        "system": "You are an academic assistant assessing student diagram submissions.",
        "user": (
            "Evaluate the Use Case Diagram image provided below.\n\n"
            "Grading scale:\n"
            "A = 95-100 %, B = 90-94 %, C = 85-89 %, D = 80-84 %, E = 70-79 %, Fx = 0-69 %\n\n"
            "Score must be a float between 0.00000 and 1.00000 rounded to five decimal places.\n\n"
            "Evaluation criteria:\n"
            "- Are core actors visible and labeled ?\n\n"
            "- Are key functional use cases present ?\n\n"
            "- Are use case names action-oriented (starting with verbs)?\n\n"
            "- Is the overall structure logically readable and reasonably organized?\n\n"
            "You are working from an image, so estimate best-effort based on visible structure.\n\n"
            "Do not penalize for fine UML syntax issues. Focus on clarity and coverage.\n\n"
            "Return a JSON object with keys 'score' and 'grade'."
        ),
        "data": row["submission_image_base64"]
    }
}
```

- almost perfect (0.81 – 1.00).

Substantial agreement was identified between human graders across all tasks using unweighted Cohen's Kappa (Kappa use case = 0.73; Kappa class = 0.69; Kappa activity = 0.70). This score does not meet the almost

An image-based input simulated a common LMS export scenario where the submission is in visual form only. The instructions emphasized visual clarity, presence of key actors/use cases, and general readability. Fine UML syntax was deemphasized in favor of holistic

interpretation.

Prompt 2 - Leverages structured inputs for more rigorous rubric-based evaluation, aligning with the course's formal expectations.

stereotypes). Additionally, the prompt introduced conditional logic (e.g., leniency for foreign students).

While Prompt 1 deliberately used image-based inputs to simulate a realistic LMS export scenario, these were later identified as suboptimal

```
ucd_prompt2 = {
  "model": "gpt-o1",
  "input": {
    "system": "You are an academic assistant assessing student diagram submissions.",
    "user": (
      "Evaluate the Use Case Diagram provided in XML format below.\n\n"
      "Grading scale:\n"
      "A = 95-100 %, B = 90-94 %, C = 85-89 %, D = 80-84 %, E = 70-79 %, Fx = 0-69 %\n"
      "Score must be a float between 0.00000 and 1.00000 rounded to five decimal places.\n\n"
      "Evaluation criteria:\n"
      "- Coverage: Are the major functionalities of a car rental system modeled? (e.g., reservation, cancellation, pickup, return, payment)\n"
      "- Actors: Are actors realistic and properly identified (e.g., Customer, Manager)?\n"
      "- Use case structure: Are relationships such as <<include>> and <<extend>> used where appropriate?\n"
      "- Naming and layout: Are use cases named clearly and concisely, in infinitive form (e.g., 'Make Reservation')?\n\n"
      "Return only a JSON object with two keys: 'score' and 'grade'."
    ),
    "data": row["submission_xml"]
  }
}
```

The XML export of the diagram allowed for the creation of a prompt that included more structured evaluation criteria targeting semantic coverage, syntax (e.g., <<inclu>, <<exten>), and naming conventions.

Prompt 3 - Emulates a more granular and realistic grading protocol, suitable for institutional standardization and fairness considerations.

due to poor layout consistency and lack of semantic richness. The inclusion of Prompt 1 thus served as a baseline for evaluating model performance under minimal structural guidance. However, Prompts 2 and 3 exclusively relied on structured XML representations, which preserved the hierarchical and semantic properties of UML diagrams and enabled more rigorous, rubric-aligned evaluation.

While the final output to all three variants included both a letter

```
ucd_prompt3 = {
  "model": "gpt-o1",
  "input": {
    "system": "You are an academic assistant assessing student diagram submissions.",
    "user": (
      "Evaluate the Use Case Diagram of a Car Rental System provided in XML format using the following rubric:\n\n"
      "- Formal and content correctness (15+ use cases, named in infinitive, correct actors): 4 pts\n"
      "- Proper use of <<include>>, <<extend>> relationships and their relevance: 3 pts\n"
      "- Use of stereotypes to improve clarity (e.g. <<actor>>, <<system>>): 2 pts\n\n"
      "Grading scale:\n"
      "A = 95-100 %, B = 90-94 %, C = 85-89 %, D = 80-84 %, E = 70-79 %, Fx = 0-69 %\n"
      "Score must be a float between 0.00000 and 1.00000 rounded to five decimal places.\n\n"
      "NOTE: If the student has 'N = 1' (foreign student), skip deductions for imperfect naming. Focus only on completeness and relevance of use cases.\n\n"
      "Return JSON with keys 'score' and 'grade'."
    ),
    "data": {
      "xml": row["submission_xml"],
      "N": 1
    }
  }
}
```

The third prompt, also with XML-based input, defined point-based subcriteria, giving precise weight to different features (e.g., 4 points for completeness, 3 points for proper relationships, 2 points for use of

grade ('A' – 'Fx') and a numerical score, the numerical value was deliberately defined as a continuous float ranging from 0.00000 to 1.00000. This decision was pedagogically and analytically motivated. In contrast to fixed-point rubrics or binary checklists, UML diagram assessment

often involves interpretive judgment across multiple overlapping criteria (e.g., semantic correctness, structural clarity, use of relationships). During human grading, even expert graders showed substantial variation in how they weighted individual flaws or strengths, especially in ambiguous or partially correct diagrams. Rather than forcing the model into rigid rubric adherence, a soft-scaled evaluation was opted for, allowing the model to express its interpretation of submission quality on a continuous spectrum. The float score thus acts as a probabilistic signal, reflecting how “complete,” “correct,” or “well-formed” the diagram appears from the model’s perspective. Letter grades were then mapped from these scores using the official university grading scale (e.g., ‘A’ = 0.95–1.00, ..., ‘F’ = below 0.70). This format mirrors real-world grading challenges in software modeling courses, where numerical objectivity is often illusory, and pedagogical consistency requires judgment rather than precision. By using continuous values, model-human agreement was later analyzed not only in grades, but also in relative rank-ordering and threshold sensitivity, offering a more nuanced view of GenAI’s evaluative behavior.

Although this evaluation was repeated for all three UML diagram types, only the Use Case Diagram prompts are shown here to illustrate the general prompting logic. Nevertheless, the same principles, structure, and scoring logic were consistently applied to the Class Diagram and Activity Diagram prompts. Each of the nine prompt configurations (3 per diagram type) was executed on the same dataset, ensuring cross-prompt comparability and enabling later analysis of grading consistency across input modalities and task complexity. The use of multiple prompts was motivated by exposing the model to different levels of structure, detail, and interpretive freedom. The aim was to identify not only its performance variability but also its potential as a robust, adaptive evaluator in real-world assessment settings.

3.3. Data preparation

As part of the data preparation phase, a final structured dataset comprising over 1800 unique student submissions collected over two academic years was compiled. Each record in the dataset represents one

student and integrates performance data, feedback content, feedback evaluations, and grading information across multiple stages of the course. The dataset was compiled into a structured matrix with **68 variables per student**, enabling a comprehensive analysis of both learning outcomes and feedback quality. Each student entry includes an anonymized **student ID**, ensuring the protection of personal data and compliance with ethical research standards. An additional categorical variable, **A**, indicates which specific version of the assignment (labeled A–E) the student completed, allowing for topic variation while maintaining task equivalence. Additional metadata includes the academic year (**AY**) in which the data was collected (2023/2024 or 2024/2025) and a binary variable **N**, which indicates whether the student was international (1) or domestic (0). This distinction was relevant due to the high proportion of foreign students (e.g., from Ukraine, Russia, Kazakhstan, Hungary, Serbia and the Czech Republic) and the potential influence of language background on feedback interpretation. However, to ensure privacy and avoid bias, no information about the nationality beyond this flag was retained or used in the analysis. Student performance on six theoretical quizzes is represented by twelve variables—six scores (**A1S – A6S**) and six derived grades (**A1G – A6G**)—corresponding to auto-evaluated knowledge checks completed throughout the semester. Similarly, four main modeling assignments (denoted as **SOU1** to **SOU4**) are included, each corresponding to a specific UML modeling task. For each assignment, the dataset stores the AI-generated feedback, as well as additional fields such as the numerical score (**SOU1S – SOU4S**),

Table 1

Summary of two-way ANOVA results across assignments.

Assignment	Feedback source p-value	Category p-value	Source vs. category p-value
SOU1	0.0404*	0.0003***	0.2016
SOU2	0.0082**	0.0462*	0.1598
SOU3	0.0271*	0.0000***	0.1566
SOU4	0.3124	0.7012	0.5345

Note: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

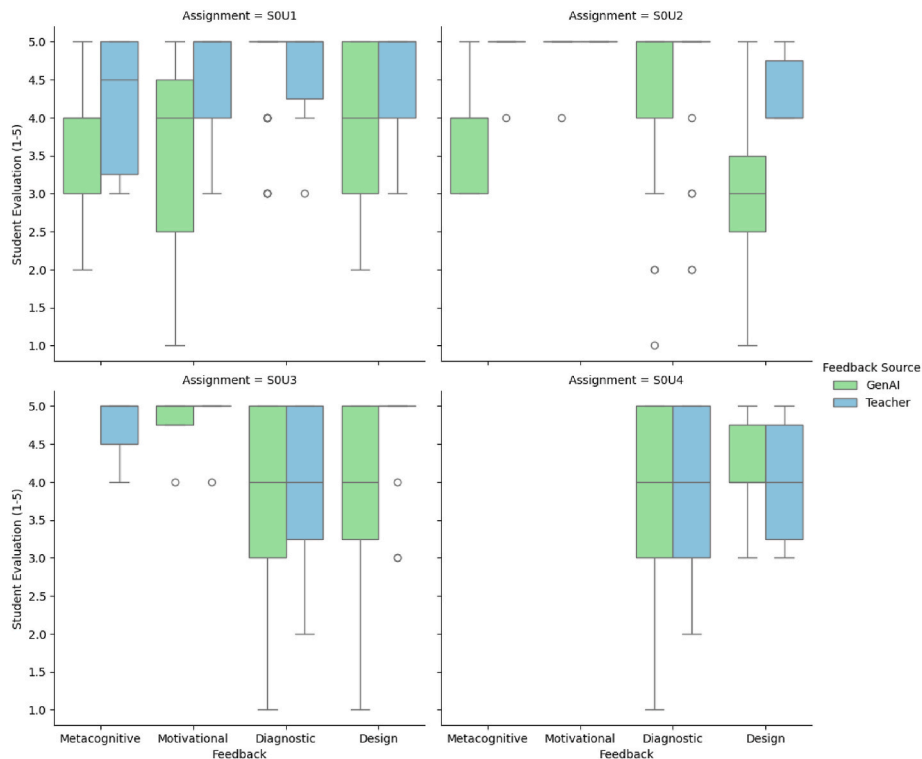


Fig. 1. Student evaluation by feedback category, source, and assignment.

a letter grade mapped from the score (S0U1G – S0U4G), a Likert-scale student rating of feedback usefulness (S0U1F – S0U4F), and a manually assigned feedback category (S0U1C – S0U4C) based on the dominant pedagogical function of the feedback comment. Each student also completed a final theoretical exam, recorded in the dataset as E (exam score) and EG (the corresponding grade based on the university grading scale). For each UML task in the final exam, two human graders independently assigned both a numeric score and a letter grade. These values are recorded in variables such as E1S_T1 and E1G_T1 (score and grade from Teacher 1 for Task 1), E2S_T2, E3G_T2, and so on, covering all combinations of tasks (E1–E3) and graders (T1, T2).

In parallel, the same diagrams were evaluated by the GPTo1 model using three distinct prompting strategies. For each of the three UML diagram types (a = use case, b = class, c = activity), and each of the three prompts (P1, P2, P3), the dataset contains both a predicted letter grade and a corresponding floating-point score ranging from 0.00000 to 1.00000. These values are stored in fields such as GPTo1P1a and GPTo1P1aS, indicating the model's grade and score for Task a using Prompt 1. This results in 18 variables in total, capturing all AI-based evaluations across prompt and task variations. This data structure enables fine-grained comparative analysis of grading alignment between humans and GenAI under different task conditions and prompt configurations. The anonymized dataset, along with variable descriptions and selected prompt examples, is available at anonymized, supporting transparency and reproducibility of the study.

4. Results

The dataset obtained has offered various insights into the feedback quality analysis. All analyzed data variables did not meet the requirements for normal distribution ($p < 0.05$) and were analyzed using non-parametric tests. The results section is organized by research questions (RQ1–RQ4).

4.1. Student-perceived helpfulness and pedagogical value of feedback (RQ1)

First, the students' evaluation of feedback was examined, where the students assigned each feedback score using a five-point Likert scale without the knowledge whether the feedback was produced by GenAI or teacher. For each assignment, a Mann-Whitney U test was used to

determine whether students rated teacher-generated and GenAI-generated feedback differently. The results showed that in the case of S0U1 (requirements specification) and S0U4 (activity diagram), the students did not find any significant qualitative differences between GenAI and the teacher ($p > 0.05$). On the other hand, statistically significant differences were identified between feedback sources in the assignment S0U2 ($p < 0.01$) and S0U3 ($p < 0.05$). A closer look at the student scores of feedback showed that in the case of S0U2 (use case diagram) and S0U3 (class diagram), they considered the teacher feedback of higher quality based on the mean score (S0U2 teacher = 4.59; S0U2 GenAI = 4.14; S0U3 teacher = 4.46; S0U3 GenAI = 4.03). Across all assignments, the students considered both feedback sources with high scores (GenAI mean = 4.09; Teacher mean = 4.38), which shows that they evaluated the feedback as beneficial regardless of the source. The lowest scores were given only to GenAI-generated feedback. Using the four feedback categories defined in RQ2 (diagnostic, design, metacognitive, motivational), boxplots of the feedback categories distributions based on student evaluations using the Likert scale for each assignment were created to confirm the observed patterns visually (Fig. 1). Assignment S0U4 did not contain any metacognitive or motivational feedback, as the activity diagram student solutions contained multiple errors. Therefore, the teachers focused more on correcting them, and the feedback was diagnostic and design-oriented. Interestingly, GenAI was not prompted to target specific feedback category, and as can be seen, it also focused only on diagnostic and design feedback in this assignment.

To examine the relationship between student feedback ratings, feedback source (GenAI or teacher), and feedback category, a two-way ANOVA was performed for each assignment (Table 1). The results confirmed a significant main effect of feedback source in S0U1–S0U3 ($p < 0.05$), where teacher feedback was consistently rated higher than GenAI feedback. Additionally, the main effect of feedback category was significant in all three assignments, indicating some feedback categories were more favorably received. However, interaction effects (Source $\times$ Category) were not statistically significant ($p > 0.05$), suggesting that the difference between GenAI and teacher feedback ratings was consistent across feedback categories. In the case of the first assignment, the students valued the highest diagnostic feedback (mean score = 4.67), while in the case of S0U2 and S0U3, students gave the highest scores to motivational feedback (mean score = 4.83; 4.89).

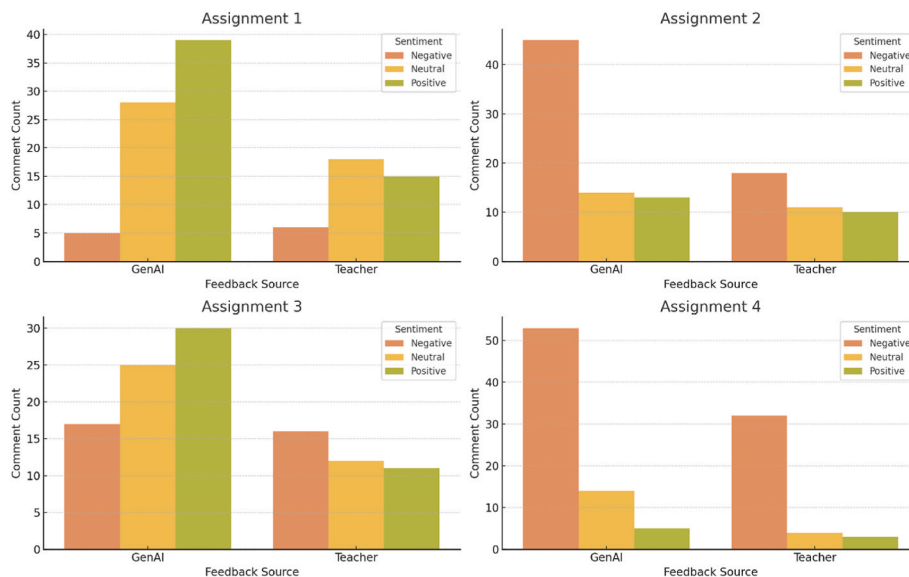


Fig. 2. Sentiment comparison of teacher and GenAI feedback.

4.2. Spontaneous feedback characteristics and type distribution (RQ2)

The GenAI model was instructed to provide formative feedback, but the prompt did not request any particular feedback type, tone, or category; thus, category diversity reflects spontaneous output. In addition to student perception, teacher-generated categorizations of feedback were analyzed. The feedback was categorized into four categories (diagnostic, design, metacognitive, and motivational) by a human evaluator. The goal was to look at the capabilities of GenAI to generate various feedback styles without specifying the prompt or requirements. The teacher and GenAI feedback were comprised mainly of diagnostic feedback type (GenAI: 66%; teacher: 48%). The other categories were evenly distributed in the case of teacher feedback (design: 19%; metacognitive: 18%; motivational: 15%). However, in the case of GenAI feedback, motivational category was the least frequent (4%) together with metacognitive (10%). GenAI was more focused on producing diagnostic (66%) and design (20%) feedback categories. The Chi-square test of independence showed that from the evaluator's perspective, both GenAI and teacher feedback were classified similarly in terms of their category ($p > 0.05$) despite descriptive differences indicating GenAI skewed more toward diagnostic feedback.

To assess the tone and character of feedback, a post-hoc sentiment analysis was conducted (Fig. 2). Traditional models such as TextBlob or VADER proved insufficient due to the use of formal Slovak and domain-specific terminology. Therefore, a custom heuristic method was adopted. This involved mapping frequent phrases to sentiment categories using a curated Slovak lexicon of pedagogically relevant terms. Sentiment was categorized as positive (e.g., “výborne” (perfect), “správne riešenie” (correct), “kompletné riešenie” (complete)), neutral (e.g., “zadanie bolo splnené” (assignment satisfied), “riešenie je v poriadku” (OK)), or negative (e.g., “chýba” (missing), “nesprávne” (wrong), “nedostatočné” (insufficient)). This enabled quantification of how the feedback tone varied between AI and instructor sources and across assignments of increasing difficulty. The goal of this analysis was to observe how feedback tone evolves across task types and sources. Fig. 2 illustrates the distribution of sentiment in feedback generated by a GenAI model and a human instructor across four student assignments of increasing complexity. This analysis served two purposes (a) to detect differences in affective expressiveness between teacher and GenAI-generated feedback, and (b) to quantify how sentiment profiles evolved across assignments of increasing difficulty and abstraction.

GenAI feedback was predominantly positive or neutral, with very limited critical sentiment in the first assignment focused on functional requirements. This suggests that the model tended to affirm student work unless explicit rubric violations were detected. In contrast, the teacher maintained a more balanced sentiment distribution even in this lower-complexity task, incorporating both encouragement and light critical cues. This pattern indicates early evidence of greater pedagogical depth in human-generated feedback. A significant shift appears in the second assignment, where students were tasked with designing use case diagrams. GenAI became notably more critical, with negative sentiment far outweighing neutral or positive tones. This change likely reflects the model's reliance on rigid evaluation criteria related to actor

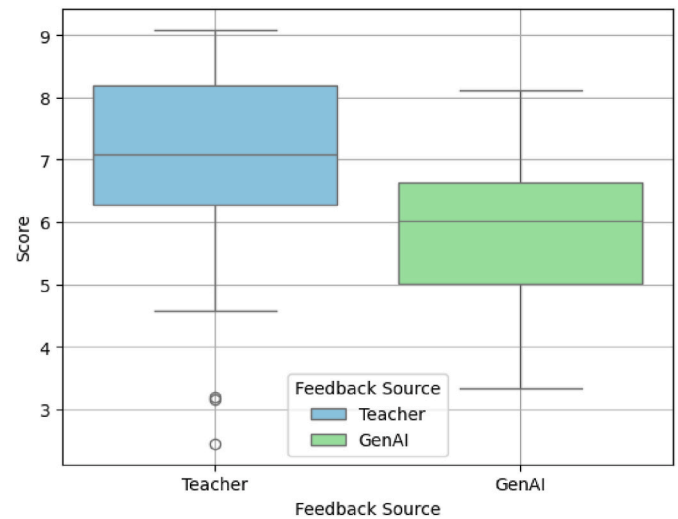


Fig. 3. Average quiz score according to feedback source.

identification, relationship stereotype (e.g., <<includ>>, <<exten>>), and naming conventions. However, the teacher's feedback remained more balanced, offering critiques that were likely contextualized and more targeted to the nature of individual submissions.

This contrast reveals a key limitation of the model — its reduced capacity for nuance when encountering partial or ambiguous diagrammatic content. As the tasks increased in abstraction with the class diagram assignment, GenAI returned to a predominantly positive tone, offering less critique despite the increased cognitive demands of modeling object-oriented systems. One interpretation is that the model may have encountered semantic limitations in evaluating inheritance structures, associations, or cohesion principles from the diagram exports (provided in XML format). In contrast, the teacher provided more negatively-coded feedback, pointing to detailed structural or conceptual issues likely overlooked by the model. This disparity reinforces the idea that the teacher was engaged in deeper evaluation aligned with pedagogical goals, whereas GenAI's assessments may have become more superficial under rising semantic complexity.

In line with these findings, GenAI returned to overwhelmingly negative sentiment in the final and most complex assignment, focused on modeling system behavior through activity diagrams. This observation is further supported by the fact that the majority of feedback in this task was classified as diagnostic. It means it was aimed primarily at identifying errors and deficiencies (see Fig. 1). The sharp rise in critical tone suggests that the model applied strict structural logic, penalizing deviations from expected flow or misuses of UML constructs. However, the human instructor remained more consistent in sentiment, continuing to provide criticism where necessary but maintaining a distribution that also included neutral and occasionally positive commentary. This divergence suggests that while GenAI enforces formalism rigidly, the instructor more carefully differentiates between critical flaws and areas for development, thus preserving a more constructive tone even under

Table 2

Linguistic metrics summary for each assignment.

	SOU1		SOU2		SOU3		SOU4	
	GenAI	Teacher	GenAI	Teacher	GenAI	Teacher	GenAI	Teacher
TTR	0.208	0.325	0.183	0.324	0.215	0.366	0.215	0.255
Hapax	0.097	0.194	0.095	0.193	0.115	0.220	0.119	0.145
Entropy	7.984	8.513	8.612	8.867	8.150	8.841	9.437	9.144
Verb Distance	6.607	5.749	7.748	7.002	7.913	6.363	7.106	6.223
Activity	0.440	0.505	0.486	0.513	0.401	0.484	0.526	0.593
Descriptivity	0.560	0.495	0.514	0.487	0.599	0.516	0.474	0.407
Thematic concentration	0.160	0.143	0.115	0.088	0.164	0.117	0.039	0.037

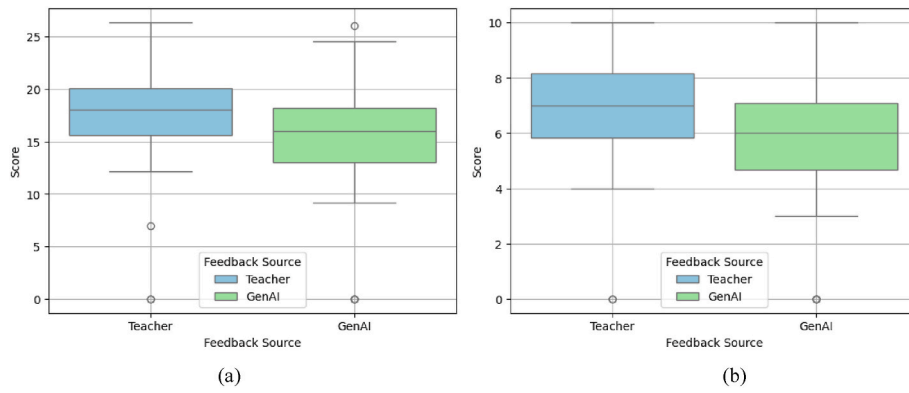


Fig. 4. Average theoretical exam (a) and final exam (b) score according to feedback source.

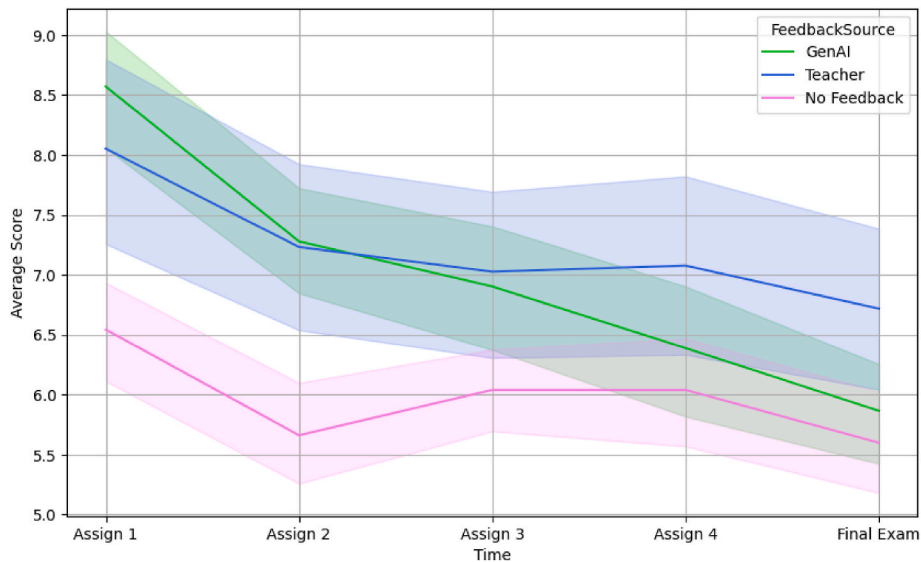


Fig. 5. Student score progression over time by feedback source.

suboptimal student performance. Overall, the results revealed that GenAI sentiment varies significantly by task type - overly optimistic in early-stage textual tasks and overly punitive in complex visual modeling. By contrast, the teacher's feedback demonstrates a more stable emotional range, flexibly adapting tone to both task demands and the quality of student work. This interpretive pattern supports earlier findings that while GenAI can deliver high-volume feedback on a scale, its pedagogical judgment remains brittle, particularly when required to reason through ambiguity, abstraction, or design variation. Accordingly, these findings suggest a strong case for hybrid feedback architectures, where GenAI operates as a first-pass evaluator and instructors retain a formative oversight role to ensure tone, relevance, and conceptual clarity.

A further linguistic analysis of feedback differences was done using QuitaUp metrics for Slovak (Cvrček et al., 2020). The results (Table 2) showed that GenAI feedback is more repetitive, while the teacher feedback is more varied and linguistically richer (higher values of TTR, Hapax and Entropy). This confirms that the teacher feedback likely contributed to better student outcomes. Descriptivity is the reverse value of activity, which is represented by the ratio of verbs and adjectives in the texts. Higher descriptivity values show that GenAI was more descriptive. Thematic concentration was higher in GenAI feedback, reflecting greater repetition of specific terms, which may explain its lower pedagogical richness.

4.3. Influence of GenAI feedback on modelling skills and performance (RQ3)

Subsequently, the impact of feedback on learning outcomes was examined. Since the final exam consisted of three UML diagram design tasks and the students had four assignments during the semester, score gain was computed as the difference between the average assignment scores and the final exam tasks. Spearman correlation showed no significant relationship between student-rated feedback quality and score gain ($p > 0.05$). Although feedback was perceived as beneficial, it did not directly impact their final score. However, a statistically significant difference in score gain was identified between feedback sources ($p < 0.05$). The students receiving teacher feedback gained on average 0.87 points more than those receiving GenAI feedback. This indicates that teacher feedback, although sometimes similarly rated to GenAI, was more effective in supporting student learning.

Additionally, students completed six automatically evaluated quizzes during the term, focused on theoretical knowledge. Teacher feedback recipients scored significantly higher on average in these quizzes ($p < 0.01$) than students with GenAI feedback (Fig. 3). This suggests that students were not only engaged with the feedback but also applied it to enhance their theoretical understanding during the term.

On the other hand, feedback source did not significantly influence ($p > 0.05$) final theoretical exam scores (Fig. 4a). This may be influenced by the overall highly rated quality of GenAI feedback or students'

increased exam preparation efforts, as presented by (Pecuchova & Drlik, 2024). Nonetheless, feedback source significantly influenced performance in the visual modeling tasks of the final exam (Fig. 4b). Statistically significant differences ($p < 0.05$) were identified between the feedback sources in the case of the average scores obtained for the three final practical tasks, where teacher feedback led to better outcomes. This suggests that while GenAI feedback was well-received, teacher feedback was more effective in helping students master abstract modeling tasks.

Further analysis showed that perceived feedback quality (average feedback student score), rather than feedback source alone, significantly predicted final exam performance ($p < 0.05$). However, the feedback source itself (GenAI or teacher) is not predictive once accounting for perceived feedback quality, theoretical exam test, and quizzes ($p > 0.05$). Mediation analysis was used to test whether the perceived feedback quality mediates the influence of feedback source on final exam performance. The analysis revealed that feedback source influenced perceived feedback quality ($\beta = -0.28$), though this effect was marginally non-significant ($p = 0.056$). The negative impact of GenAI feedback on performance was mediated mainly through perceived quality rather than the source itself. When adding theoretical exam test and quizzes results, perceived feedback quality remained a significant predictor of final exam outcomes. However, the indirect effect of feedback source on performance was reduced from -0.30 to -0.13 , which means that theory and practical scores explain a substantial part of final exam variance, as expected. Bootstrapping with 5000 resamples confirmed a small but significant indirect effect of feedback source on final exam performance through perceived feedback quality (indirect effect = -0.13 , 95% CI $[-0.31, -0.0006]$).

Regression analysis of feedback categories showed that most categories did not significantly impact student performance ($p > 0.05$), except for diagnostic feedback. Overall, the motivation behind the classification of the feedback into the four categories was to examine whether GenAI can, without a specific prompt, generate not only diagnostic but also other variants of feedback. This was confirmed as GenAI created 11% metacognitive and 5% motivational feedback. Specifically, GenAI-generated diagnostic feedback significantly reduced score gains by 0.47 points compared to teacher-generated diagnostic feedback ($p < 0.05$).

A mixed linear regression model was used to analyze progression across assignments (S0U1 - S0U4) and whether feedback source (GenAI or teacher) influenced learning advancement. The results (Fig. 5) include a control group of students who received no feedback. The experiment was done during two terms, where on the first year the students had the assignments only assessed using a score, and another year, the students also obtained textual feedback from the teacher or GenAI. The analysis (Fig. 5) confirmed that feedback had a significant positive effect on learning, with teacher feedback leading to greater improvement than GenAI feedback. The first assignment (S0U1), which focused on functional and non-functional requirements, had the highest average score, while subsequent assignments required visual modeling and posed greater challenges. Although average scores declined over time, feedback helped mitigate this decline, particularly for the teacher feedback group. The results suggest that while GenAI feedback supports learning to some extent, it is less effective than teacher feedback. A hybrid approach, where the teacher would post-edit the GenAI feedback, may help balance effectiveness with reduced workload and offer more insights for students on their assignments in a more timely manner. On the other hand, the results showed that GenAI models still lag behind humans and cannot produce complex and beneficial formative feedback. Nevertheless, any type of feedback improves the learning performance and helps students better understand the abstract diagrammatic visual tasks. The high student ratings confirmed the beneficial influence of feedback.

4.4. Automated grading of UML diagrams based on XML and prompt strategy (RQ4)

GenAI model GPTo1 was used to generate not only feedback but also to score each final exam task. The final exam consisted of three UML diagrammatic visual tasks: (a) use case diagram, (b) class diagram, and (c) activity diagram. The model was prompted using three distinct prompt strategies, and scores were obtained for each task and student solution. GenAI-generated scores were mapped to grades based on the course curriculum. The evaluation was aimed at analyzing the performance of GPTo1 in the automated assessment of diagrammatic visual tasks and, together with feedback generation, could offer a fully self-

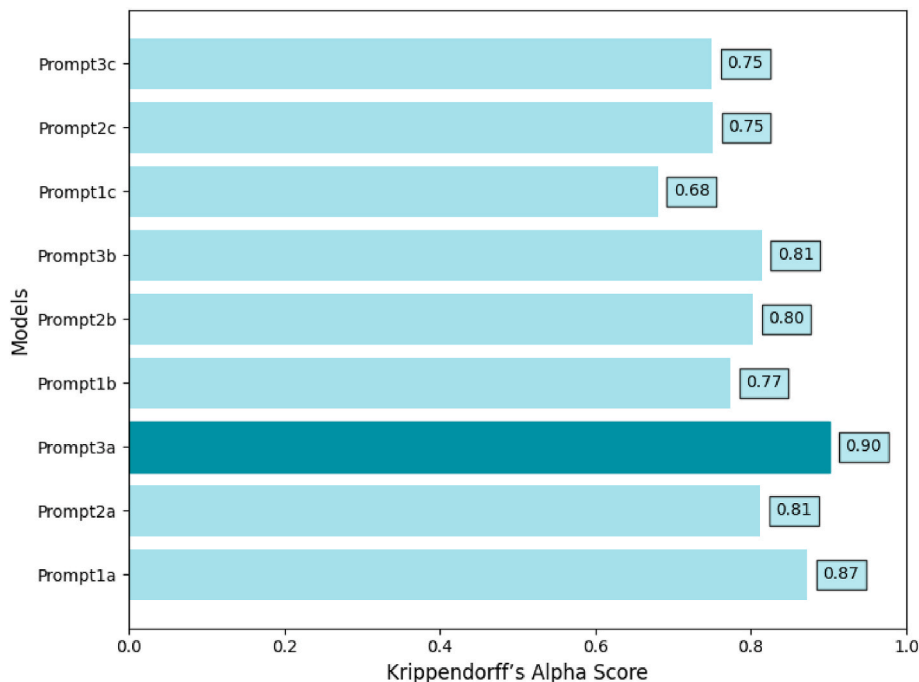


Fig. 6. Krippendorff's Alpha Agreement Scores using Different Prompt Strategies for Final Exam Tasks.

Table 3

A summary of statistically significant and non-significant results.

Experiment/ Measure	Statistical Test	Significance	Finding
Student Perception (SOU1)	Mann-Whitney U	$p > 0.05$	No significant difference (Teacher vs GenAI)
Student Perception (SOU2)	Mann-Whitney U	$p < 0.01$	Teacher feedback rated higher (Mean 4.59 vs 4.14)
Student Perception (SOU3)	Mann-Whitney U	$p < 0.05$	Teacher feedback rated higher (Mean 4.46 vs 4.03)
Student Perception (SOU4)	Mann-Whitney U	$p > 0.05$	No significant difference (Teacher vs GenAI)
Feedback Category Effect (SOU1)	Two-way ANOVA	$p < 0.001$	Diagnostic feedback is most valued
Feedback Category Effect (SOU2)	Two-way ANOVA	$p < 0.05$	Motivational feedback is most valued
Feedback Category Effect (SOU3)	Two-way ANOVA	$p < 0.001$	Motivational feedback is most valued
Feedback Category Effect (SOU4)	Two-way ANOVA	$p > 0.05$	No significant category effect
Score Gain (Assignments > Final Exam)	Non-parametric test	$p < 0.05$	Teacher +0.87 points > GenAI
Quiz Performance	Non-parametric test	$p < 0.01$	The teacher group scored higher
Final Exam - Visual Modeling Tasks	Non-parametric test	$p < 0.05$	Teacher feedback led to higher outcomes
Final Exam - Theoretical Exam	Non-parametric test	$p > 0.05$	No significant difference by feedback source
Mediation Analysis	Bootstrapped regression		A small but significant indirect effect of the feedback source on the final exam
Diagnostic Feedback Effect	Regression	$p < 0.05$	GenAI diagnostic feedback reduced score gain by -0.47
Progression Across Assignments	Mixed linear regression	$p < 0.05$	Teacher feedback > GenAI > No feedback
GenAI Grading - Use Case Diagrams	Fleiss' Kappa	$Kappa = 0.674$	Substantial agreement with Prompt 3
GenAI Grading - Class Diagrams	Fleiss' Kappa	$Kappa = 0.631$	Substantial agreement with optimized prompt
GenAI Grading - Activity Diagrams	Fleiss' Kappa	$Kappa = 0.600$	Borderline substantial agreement
GenAI Grading - Use Case Diagrams	Krippendorff's Alpha	$Alpha = 0.901$	Almost perfect agreement with Prompt 3
GenAI Grading - Class Diagrams	Krippendorff's Alpha	$Alpha = 0.814$	Almost perfect agreement with Prompt 3
GenAI Grading - Activity Diagrams	Krippendorff's Alpha	$Alpha = 0.664$	Substantial agreement

sufficient system. The estimated grades were compared with the human grader's assessments. This comparison had almost perfect agreement for all three tasks (QWK a = 0.91; QWK b = 0.92; QWK c = 0.95). This way, it was possible to compare the output of GPTo1 and evaluate its capability to assess UML diagrams.

Fleiss' Kappa and Krippendorff's Alpha measured agreement between GPTo1 assessments and human graders. Both measures focus on ordinal data, where Fleiss' Kappa does not consider the level of disagreement and evaluates the performance strictly. Substantial agreement was achieved for use case diagrams across all prompts, with the third prompt

performing the best (Fleiss' Kappa: P1 = 0.621; P2 = 0.622; P3 = 0.674). Class diagram assessment improved with prompt optimization, reached substantial agreement (Fleiss' Kappa: P1 = 0.562; P2 = 0.625; P3 = 0.631). The third task exhibited lower agreement, but improvement in agreement was observed with different prompt strategies, where borderline substantial agreement was achieved using the third prompt strategy (Fleiss' Kappa: P1 = 0.527; P2 = 0.558; P3 = 0.600).

Krippendorff's Alpha offers a less strict approach to the agreement. Differences were observed between various prompt strategies, where two of the diagrams were evaluated at almost a perfect agreement level (Fig. 6). The highest agreement was obtained for use case diagram assessment using the third prompt (Alpha P3 = 0.901). The second prompt showed a significantly lower assessment performance compared to the first and third prompts (Alpha: P1 = 0.873; P2 = 0.812). In the case of the class diagram assessment, improvement was identified with the second and third prompts, moving from substantial to almost perfect agreement (Alpha: P1 = 0.774; P2 = 0.802; P3 = 0.814). The third task achieved a considerable level of agreement with human graders, with significant improvement in the second and third prompts. However, with only slightly higher performance in favor of the second prompt strategy (Alpha: P1 = 0.631; P2 = 0.664; P3 = 0.663).

The results confirmed that GPTo1 can assess UML diagrams with a substantial level of agreement. Moreover, for specific diagram types, such as use case diagrams, it can reach almost perfect agreement. Overall, prompting strategies indicate improvement in assessment performance without additional model training. Table 3 summarizes the statistically significant and non-significant results, which will be discussed in more detail in the discussion section, together with the implications of this research for different domains and stakeholders.

5. Discussion

This discussion interprets the findings in relation to the four research questions, focusing on student perceptions of feedback, the pedagogical characteristics of feedback, learning outcomes, and the reliability of GenAI-supported grading. Rather than reiterating empirical results presented in Section 4, the discussion aims to contextualize the observed patterns within existing research on formative feedback, diagrammatic modeling, and hybrid human-AI educational systems.

RQ1: How can the student-perceived helpfulness and pedagogical value of GenAI-generated feedback be compared to the instructor's feedback across UML modeling tasks?

With respect to RQ1, the findings indicate that students generally perceived both teacher-generated and GenAI-generated feedback as helpful, confirming that formative feedback is valued in complex UML modeling tasks regardless of its source. However, statistically significant differences emerged in assignments involving higher semantic and structural complexity, where teacher feedback was consistently rated higher. This suggests that while GenAI feedback can satisfy baseline expectations for usefulness, its perceived value decreases as tasks demand deeper interpretive judgment, design rationale, and contextual understanding. The absence of significant differences in some assignments further indicates that task characteristics mediate how feedback sources are perceived. In less abstract or more procedurally constrained tasks, GenAI feedback may be experienced as equally valid, whereas in conceptually rich modeling tasks, students appear more sensitive to qualitative depth and pedagogical nuance.

RQ2: What types of feedback does GenAI produce spontaneously without explicit prompt conditioning, and how does this distribution compare to that of teacher-generated feedback?

Regarding RQ2, the analysis of feedback categories revealed that both GenAI and teacher feedback addressed diagnostic and design-

oriented aspects of student work, while motivational and metacognitive feedback appeared less frequently and predominantly in teacher-generated responses. The perceived value of feedback categories varied across assignments, with diagnostic feedback most valued in specification-focused tasks and motivational feedback becoming more prominent in complex UML design assignments. These patterns suggest that students' feedback preferences shift alongside increasing task complexity and cognitive demand. Importantly, the limited and unsystematic emergence of motivational and metacognitive feedback in GenAI outputs highlights a key limitation of unprompted AI-generated feedback. Without explicit instructional guidance embedded in prompt design, GenAI tends to prioritize evaluative commentary over dialogic or reflective engagement, reinforcing earlier concerns that automated feedback systems may struggle to support higher-order learning processes unless pedagogically framed.

RQ3: Does receiving GenAI feedback influence students' modelling skills and improve their performance on subsequent modeling tasks and final exams, relative to students receiving teacher or no feedback?

In relation to RQ3, the results demonstrate that feedback source significantly influenced learning outcomes in applied visual modeling tasks. Students receiving teacher feedback showed greater score gains across assignments, higher quiz performance, and stronger performance in final exam modeling tasks compared to those receiving GenAI feedback. Mediation analysis further indicated that perceived feedback quality played a central role in explaining these differences, suggesting that how students interpret and trust feedback is more consequential than the feedback source itself. Notably, GenAI-generated diagnostic feedback was associated with reduced score gains compared to teacher diagnostic feedback. This pattern is consistent with the linguistic analysis, which showed that GenAI feedback exhibited higher repetition and lower lexical diversity, while teacher feedback contained more specific reformulations, examples, and task-oriented guidance—features that are more likely to support actionable uptake and learning transfer. While the magnitude of this effect may appear modest, its cumulative impact becomes relevant in scenarios where GenAI feedback is deployed intensively or substitutes human input. Longitudinal analysis confirmed that both teacher and GenAI feedback supported learning progression relative to no-feedback conditions, yet teacher feedback consistently produced stronger gains. Together, these findings suggest that GenAI feedback, in its current form, functions more effectively as a supplementary support rather than a replacement for human instructional feedback in complex modeling contexts.

RQ4: Can GenAI accurately grade visual UML diagrams based on XML representations with agreement levels comparable to expert human graders, and how do different prompt strategies influence this grading performance?

Finally, addressing RQ4, the findings show that GenAI achieved substantial to almost perfect agreement with human graders when evaluating UML diagrams, particularly under optimized prompting conditions. This level of agreement suggests that GenAI can reliably approximate human grading judgments in structured modeling tasks, supporting its potential role in scalable assessment workflows. However, agreement varied across diagram types and prompt formulations, underscoring the sensitivity of GenAI evaluation to input structure and rubric specificity. Importantly, high grading agreement does not necessarily imply pedagogical effectiveness. While GenAI may perform well as an evaluator, this does not guarantee that its feedback supports conceptual understanding or learning progression, reinforcing the distinction between evaluative accuracy and instructional value. Although prompt design was not treated as an independent experimental variable, the observed variability across prompt strategies

indicates that prompting constitutes a critical design dimension that warrants systematic treatment in future methodological research. Prompt strategy served as a design factor in this study which conditioned grading agreement and feedback tone and because of that the results are interpreted under these specific prompt conditions.

5.1. Research implications

The following implications distinguish between findings directly supported by the empirical results of this study and forward-looking design propositions intended to inform future research and educational practice. This study suggests that, based on the observed patterns, GenAI has the potential to support a shift toward feedback being conceptualized from a scarce teacher resource into a scalable pedagogical infrastructure, enabling more frequent and personalized formative assessment. However, AI feedback could be embedded in hybrid teacher-AI models to maximize learning, where educators motivate students, add nuance, context, and critical thinking prompts that AI alone cannot provide.

Although this study focused on UML diagramming in software engineering, the findings may be cautiously transferable to other disciplines that involve structured, rule-based, and semantically rich tasks. However, such transferability remains hypothetical and requires empirical validation in domain-specific contexts, where this paper may serve as a starting point. The demonstrated strengths of GenAI feedback and its potential to become a cognitive partner of teachers are directly relevant to other study fields such as architecture, biology, business and economics, computer science, and mathematics, where students create formal representations that require iterative refinement. In this context, GenAI functions as a cognitive partner in the sense of distributed cognition, supporting diagnostic reasoning and iterative feedback cycles while pedagogical judgment and accountability remain with the instructor, consistent with the framing of AI as a cognitive partner in learning set-ups (Huang et al., 2022).⁷ As in UML modeling, these tasks can overwhelm teachers' capacity to provide detailed and timely feedback at scale, making them potential candidates for AI-supported augmentation. However, if the observed limitations like GenAI's tendency toward repetition, lack of nuance, and reduced effectiveness in complex or abstract assignments are considered, these findings suggest that hybrid AI-teacher feedback models may be necessary across domains rather than fully automated approaches. More broadly, the study indicates a potential shift toward feedback-abundant learning environments in disciplines grounded in structured symbolic or visual representations. Such extrapolation should therefore be considered plausible primarily for domains characterized by formal syntax, constrained solution spaces, and evaluable visual or symbolic representations.

If GenAI can reliably provide formative feedback in visual modelling tasks, curriculum design can move away from the traditional constraint of limited teachers' feedback toward models that are feedback-rich, iterative, and scalable. This may support instructional designs emphasizing iterative cycles of modelling, feedback, and revision, without being limited by teacher capacity. Such approaches could enable more personalized learning pathways and support the use of authentic, open-ended assessments that would otherwise be difficult to sustain at scale.

Deploying GenAI feedback may also lead to creation of new layer of learning analytics evidence in form of interaction logs that contain timing, frequency and type of feedback. Another evidence may present recurring error patterns in diagram design structure that can timely alert the educators to deal with these errors. When responsibly overseen, this layer can support instructors to detect systematic misunderstandings, monitor feedback consistency across prompt versions and evaluate whether students act based on the obtained feedback. This may lead to reframing GenAI not only as a feedback generator but also as a tool that supports course improvement and scalable formative assessment.

Moreover, if GenAI feedback can be deployed at scale, assessment in engineering programs could gradually support a reorientation within

the constraints identified in this study. Traditionally, assessments in engineering have been constrained by the limited capacity of teachers to provide timely, formative feedback for complex assignments or projects, resulting in heavy reliance on summative exams to validate learning. Implementing GenAI feedback, even with some limitations identified in this study, allows students to receive iterative guidance throughout the design and problem-solving process. This may support a gradual shift toward competency-based and mastery-oriented assessment models, where progression depends on demonstrated skill development over time rather than high-stakes exams.

As demonstrated by the variability in grading agreement and feedback tone across the three prompt versions tested in this study, such deployment presupposes that the prompts used to guide the AI remain stable and robust across different contexts. Based on the results, the structure and phrasing of prompts can substantially influence both the tone and accuracy of GenAI feedback. Although three prompt versions were tested for each diagram type to simulate diverse evaluative perspectives, it was observed that the model's grading behavior and style of response varied depending on the formulation and specificity of the prompt. This highlights that prompt engineering should be treated as an integral component of educational design rather than a purely technical concern. Prompts must be continually validated to ensure reliability across variations in input quality, student background, and assignment type. Without such validation, the risk of inconsistent or biased evaluation increases, potentially undermining trust in GenAI-supported feedback systems.

Simultaneously, agreement is also expressed with other researchers (Correia et al., 2017; Sandu et al., 2024; Wang et al., 2024) that study programs should adopt authentic assessments that mirror real-world practice, since the burden of detailed feedback can be divided between the teachers and their potential cognitive partner (Er et al., 2025). However, this shift requires careful consideration of the results of studies similar to this study and understanding the limitations of GenAI feedback in particular settings and environments. In this sense, AI-driven feedback may support a transition from episodic evaluation toward more continuous learning and skill refinement, rather than replacing established assessment practices outright.

Considering the pedagogical models and learning theories, the integration of AI-generated feedback has the potential to both foster and undermine models of self-regulated learning (SRL). On one hand, the immediacy and abundance of feedback can promote SRL by giving students the resources to monitor their progress, set goals, and iteratively refine their work without being constrained by instructor availability. In this role, AI may function as a scaffold for metacognitive monitoring and strategic planning. However, considering the imperfections and limitations of the GenAI feedback confirmed also in this study, an over-reliance on AI feedback may risk eroding students' autonomy and reflective capacity if students passively accept suggestions without critically evaluating their validity or integrating them into their reasoning processes. This dependency could weaken the students' ability to interpret, question, and act on feedback independently. To align AI-supported feedback with SRL models, curricula should incorporate structured opportunities for reflection and comparison between AI and teacher feedback.

The outcomes of this research can carry significant implications for multiple stakeholders in higher education. From the teachers' perspective, hybrid teacher-AI models may help reduce repetitive diagnostic feedback while preserving human contributions in areas requiring judgment, motivation, and contextualization. This redistribution of roles may allow instructors to manage larger and more diverse cohorts while maintaining instructional quality. Students may benefit from immediate and iterative feedback that supports autonomy, yet they must also develop skills to critically interpret and evaluate AI-generated feedback.

At the institutional level, GenAI feedback systems represent a strategic opportunity to scale high-quality learning support while reducing faculty workload. At the same time, they raise challenges related to

ethics, equity, governance, and accountability. For example, inconsistent interpretation of non-canonical yet conceptually valid UML structures by GenAI systems may disproportionately disadvantage students whose modeling approaches diverge from standardized patterns. Another challenge is different prompt formulation by students or teachers, where not only differences may be caused by different language input or various phrasing. Building feedback and GenAI literacy of all stakeholders should become a central component of learning in an AI-augmented educational system. University leaders must therefore take a proactive role in developing policies and investing in hybrid infrastructures. However, firstly, they need to ensure the development of a new set of AI-related professional skills among faculty staff. Subsequently, together with systematic monitoring and continual transfer of the results of contemporary educational research, the academic institutions can successfully adapt to the new reality. These implications should be interpreted in light of the study's scope, which focuses on formative feedback in UML modeling, and should not be read as prescriptive claims for universal adoption without further empirical validation.

5.2. Research limitations

While this study provides valuable empirical evidence on the role of GenAI in formative feedback for UML modelling, its several limitations should be acknowledged. The research was conducted within a single institutional and disciplinary context (software engineering at one university), which may limit the generalizability of findings to other domains or educational settings. A quasi-experimental design was applied, which is pragmatic, but relied on naturally occurring groups of unequal size (GenAI feedback $n = 72$, teacher feedback $n = 39$, without feedback $n = 151$). This decision may lead to group imbalance and uncontrolled confounding factors. The annotation of feedback categories was performed by a single human grader, raising concerns about subjectivity even though GenAI was not prompted to create specific oriented feedback. Although the study relied on a single generative model (GPTo1), which limits the generalizability of the findings to other LLMs with different architectures, training data, or fine-tuning strategies, this model is one of the most used LLMs. Integration of multiple GenAI models would require more resources and assignments and could be considered as further step in future research. In addition, several limitations are associated with the design and application of the prompts used for GenAI evaluation. While three versions of prompts were created to capture different grading perspectives, they were developed by a single teacher and thus may reflect implicit assumptions about pedagogical focus and evaluative priorities. Given the well-documented sensitivity of LLMs to prompt formulation, variations in phrasing, input structure, and rubric specificity could have influenced the model's responses. Moreover, although the model generated feedback in Slovak, the prompts themselves were written in English, potentially introducing minor interpretive mismatches. Importantly, while the prompts were designed to encourage thoughtful, formative responses, they did not include explicit directives to generate motivational or metacognitive feedback. As a result, the emergence of such feedback types was organic rather than intentional. This limits the ability to fully assess the model's capacity to produce higher-order pedagogical guidance. Finally, due to ethical and publication constraints, full versions of the original student tasks or prompt inputs could not be included in the paper, instead, representative samples were provided to illustrate the structure and intent of the prompt strategy. Despite these constraints, the prompt design enabled a structured yet flexible evaluation of AI-generated feedback under conditions that closely resemble authentic educational use. The resulting analysis offers meaningful insight into how prompting strategies influence the quality and character of AI-generated formative assessment.

6. Conclusion

This conclusion focuses on synthesizing empirically supported findings, acknowledged limitations, and clearly defined directions for future research. Presented study examined the usefulness of GenAI for providing formative feedback on UML diagramming tasks in higher education. The findings demonstrate that students perceived GenAI feedback as beneficial and, in some cases, comparable to teacher comments. However, teacher feedback proved more effective in improving performance on complex modeling tasks, quiz outcomes, and final exam results. Linguistic and sentiment analyses further revealed that GenAI feedback, while scalable and consistent, lacked the richness, nuance, and adaptability of human-provided guidance. Nevertheless, even GenAI feedback offered improvement to performance in comparison to students who did not receive any feedback. Notably, the study also showed that GenAI can achieve substantial agreement with human graders in assessing UML diagrams, particularly when prompt design is optimized, highlighting its potential as a scalable evaluation tool and as a valuable cognitive partner in education.

However, considering the limitations of the study, caution is required before generalizing the findings. Addressing these limitations in future research will be essential to understand the educational value and risks of GenAI-driven feedback fully.

This study offers a contribution to the evidence that GenAI is emerging not merely as a supplementary tool but may potentially serve as an infrastructural and essential element of future higher education systems, capable of transforming how feedback, assessment, and learning are organized. The challenge ahead lies in designing hybrid teacher-AI ecosystems that combine the scalability of AI with the pedagogical depth of human expertise. This work contributes to an emerging vision of new systems of learning, where AI reshapes the feedback landscape from a teacher-limited resource into a scalable infrastructural capability that can inform the design of hybrid feedback systems in higher education.

CRedit authorship contribution statement

Janka Pecuchova: Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Lubomir Benko:** Writing – review & editing, Writing – original draft, Visualization, Validation, Formal analysis. **Martin Drlik:** Writing – original draft, Supervision, Investigation.

Ethics statement

All procedures performed in the study were approved and in accordance with the ethical standards of the Ethics Committee of the University of Constantine the Philosopher in Nitra (approval number UKF/225/2024/191013:002). Participation in this research was voluntary and fully anonymized.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The dataset analyzed during the current study is available at

<https://github.com/J-Pecuchova/XML-based-UML-models-dataset>.

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